
Week 9: Pipeline Creation using script

1. Evaluation of Jenkins pipeline.
2. **WORKING ON BUILD TRIGGERS FOR LAST JENKINS PIPELINE**
3. Hands-on practice on creation of scripted Jenkins pipeline.
4. Take the screenshots for above task

Procedure

General :

Description :- provide the description of the project

Build triggers: here we can provide with build triggers of you choice

Advance project option:

Definition:

Choose pipeline script

Here code for pipeline -script

Copy the code there

```
pipeline {
  agent any
  tools{
    maven 'MAVEN-HOME'
  }
  stages {
    stage('git repo & clean') {
      steps {
        //bat "rmdir /s /q mavenjava"
        bat "git clone provide your github link"
        bat "mvn clean -f mavenjava"
      }
    }
    stage('install') {
      steps {
        bat "mvn install -f mavenjava"#project name#
      }
    }
    stage('test') {
      steps {
```

```
        bat "mvn test -f mavenjava"
    }
}
stage('package') {
    steps {
        bat "mvn package -f mavenjava"
    }
}
}
```

Dashboard > All > New Item

New Item

Enter an item name

Week9-Pipeline

Select an item type

-  **Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
-  **Maven project**
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
-  **Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.
-  **Multi-configuration project**

OK

Assignments x SE-LAB-3-1-IT-F59-A-2025 x Week9-Pipeline Config [Jen] x Jenkins pipeline creation x 10.11.106.195 x

localhost:8888/job/Week9-Pipeline/configure

Dashboard > Week9-Pipeline > Configuration

Configure

- General
- Build Triggers**
- Advanced Project Options
- Pipeline

Build Triggers

☐ Build after other projects are built ?

☒ Build periodically ?

Schedule ?

H 18 * * *

Would last have run at Monday, 6 October, 2025, 6:37:52 pm India Standard Time; would next run at Tuesday, 7 October, 2025, 6:37:52 pm India Standard Time.

☐ Build whenever a SNAPSHOT dependency is built ?

☒ GitHub hook trigger for GITScm polling ?

☒ Poll SCM ?

Schedule ?

H/5 * * * *

Save Apply

Assignments x SE-LAB-3-1-IT-F59-A-2025 x Week9-Pipeline Config [Jen] x Jenkins pipeline setup x se-lab/pom.xml at main · m... x

localhost:8888/job/Week9-Pipeline/configure

Dashboard > Week9-Pipeline > Configuration

Configure

- General
- Build Triggers
- Advanced Project Options
- Pipeline**

Pipeline

Definition

Pipeline script

Script ?

```
10 }
11 }
12 stage('Git Repo & Clean') {
13   steps {
14     bat "git clone https://github.com/manisai28/se-lab.git"
15     bat "mvn clean -f mavenjava"
16   }
17 }
18 stage('Install') {
19   steps {
20     bat "mvn install -f mavenjava"
21   }
22 }
23 stage('Test') {
24   steps {
25     bat "mvn test -f mavenjava"
26   }
27 }
```

☒ Use Groovy Sandbox ?

Pipeline Syntax

Save Apply

Assignments x SE-LAB-3-1-IT-F59-A-2025 x Week9-Pipeline #6 Console x Jenkins pipeline setup x se-lab/pom.xml at main · m... x

localhost:8888/job/Week9-Pipeline/lastBuild/console

Dashboard > Week9-Pipeline > #6

Console Output

Changes

Edit Build Information

Delete build #6

Timings

Pipeline Overview

Pipeline Console

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

Started by user admin

```
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\Week9-Pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Tool Install)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Welcome Message)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] echo
Welcome to Jenkins Pipeline Execution - Build #6
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
```

Assignments x SE-LAB-3-1-IT-F59-A-2025 x Week9-Pipeline #6 Console x Jenkins pipeline setup x se-lab/pom.xml at main · m... x

localhost:8888/job/Week9-Pipeline/lastBuild/console

Dashboard > Week9-Pipeline > #6

```
[INFO] --- surefire:3.2.5:test (default-test) @ E-Ticketing ---
[INFO] No tests to run.
[INFO]
[INFO] --- war:3.3.1:war (default-war) @ E-Ticketing ---
[INFO] Packaging webapp
[INFO] Assembling webapp [E-Ticketing] in [C:\ProgramData\Jenkins\.jenkins\workspace\Week9-Pipeline\se-lab\target\E-Ticketing-system]
[INFO] Processing war project
[INFO] Copying webapp resources [C:\ProgramData\Jenkins\.jenkins\workspace\Week9-Pipeline\se-lab\src\main\webapp]
[INFO] Building war: C:\ProgramData\Jenkins\.jenkins\workspace\Week9-Pipeline\se-lab\target\E-Ticketing-system.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.772 s
[INFO] Finished at: 2025-10-07T10:54:27+05:30
[INFO] -----
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Dashboard > Week9-Pipeline > #6 > Pipeline Console

Build #6

[Rebuild](#)[Overview](#)[Configure](#)[...](#)

Success 14 min ago in 55 sec

- Tool Install
- Welcome Message
- Clean Workspace & Clone Repo
- Install
- Test
- Package

```
24 [INFO] No tests to run.
25 [INFO]
26 [INFO] --- war:3.3.1:war (default-war) @ E-Ticketing ---
27 [INFO] Packaging webapp
28 [INFO] Assembling webapp [E-Ticketing] in [C:\ProgramData\Jenkins\.jenkins\workspace\Week9-Pipeline\se-lab\target\E-
Ticketing-system]
29 [INFO] Processing war project
30 [INFO] Copying webapp resources [C:\ProgramData\Jenkins\.jenkins\workspace\Week9-Pipeline\se-lab\src\main\webapp]
31 [INFO] Building war: C:\ProgramData\Jenkins\.jenkins\workspace\Week9-Pipeline\se-lab\target\E-Ticketing-system.war
32 [INFO] -----
33 [INFO] BUILD SUCCESS
34 [INFO] -----
35 [INFO] Total time: 2.772 s
36 [INFO] Finished at: 2025-10-07T10:54:27+05:30
37 [INFO] -----
```

Jenkins 2.489

Dashboard > Week9-Pipeline >

Status

Week9-Pipeline

[Edit description](#)

Changes

Build Now

Configure

Delete Pipeline

Full Stage View

Stages

Rename

Pipeline Syntax

GitHub Hook Log

Polling Log

Week 9: Scripted Pipeline for Maven project with Git integration

Stage View

		Declarative: Tool Install	Welcome Message	Clean Workspace & Clone Repo	Install	Test	Package
Average stage times: (full run time: ~29s)		165ms	328ms	10s	5s	3s	4s
#12	Oct 07 11:31 No Changes	88ms	299ms	6s	4s	4s	7s
#11	Oct 07 11:30 No Changes	139ms	261ms	7s failed	126ms failed	83ms failed	94ms failed
#10	Oct 07 11:21 No Changes	255ms	346ms	6s	4s	4s	4s

Builds

Assignments SE-LAB-3-1-IT-FS9-A Stages [Week9-Pipel Jenkins pipeline setu manisai28/se-lab New Tab

localhost:8888/job/Week9-Pipeline/multi-pipeline-graph/

Jenkins

Dashboard > Week9-Pipeline > Stages

Build Week9-Pipeline

Build Configure

id	pipeline
#17	Start Tool Install Welcome Messa... Clean Workspac... Install Test Package Jar Build End
#16	Start End
#15	Start Tool Install Welcome Messa... Clean Workspac... Install Test Package End

Apply and save